

Mineral Identification and Rock Classification

Objective

The objective of this laboratory exercise was to learn techniques for identifying common minerals and to classify rock samples as igneous, sedimentary, or metamorphic based on their mineral composition and texture. Students practiced using physical properties (hardness, cleavage, luster, color, streak) to identify minerals and applied classification schemes to determine rock types.

Background

Minerals are naturally occurring, inorganic solids with a definite chemical composition and crystalline structure. Physical properties used for identification include:

- **Hardness:** Measured on Mohs scale (1=talc, 10=diamond)
- **Cleavage:** Tendency to break along planes
- **Luster:** Appearance in reflected light (metallic, non-metallic)
- **Streak:** Color of powdered mineral
- **Color:** Can be variable for some minerals

Rocks are classified into three main groups based on their origin:

- **Igneous:** Formed from cooling of magma/lava
- **Sedimentary:** Formed from accumulation and lithification of sediments
- **Metamorphic:** Formed from transformation of existing rocks under heat/pressure

Materials and Methods

Materials:

- Mineral samples: quartz, feldspar, calcite, mica, halite, pyrite, hematite, magnetite
- Rock samples: granite, basalt, sandstone, limestone, slate, gneiss, schist
- Streak plates
- Glass plate (hardness ~5.5)
- Copper penny (hardness ~3.5)
- Fingernail (hardness ~2.5)
- Hand lens (10x magnification)
- Dilute HCl solution (for carbonate test)

Procedure:

1. For each mineral sample, observe and record:
 - Color and luster
 - Hardness (using scratch tests)
 - Cleavage or fracture
 - Streak color
 - Any other distinctive properties
2. For each rock sample:

- Identify visible minerals
- Describe texture (grain size, shape, arrangement)
- Classify as igneous, sedimentary, or metamorphic
- Determine specific rock name

Results

Mineral Identification

Mineral	Hardness	Cleavage	Luster	Streak	Other Properties
Quartz	7	None (conchoidal fracture)	Vitreous	White	Common color: clear/pink/purple
Feldspar	6	Good in 2 directions at 90°	Vitreous to pearly	White	Pink or white color common
Calcite	3	Perfect rhombohedral (3 directions)	Vitreous	White	Effervesces with HCl
Mica (Muscovite)	2-2.5	Perfect in 1 direction (sheets)	Pearly to vitreous	White	Transparent, flexible sheets
Pyrite	6-6.5	None	Metallic	Greenish-black	"Fool's gold", cubic crystals
Magnetite	5.5-6.5	None	Metallic	Black	Magnetic, black color

Rock Classification

Rock Sample	Minerals Present	Texture	Classification
Sample A	Quartz, feldspar, mica	Coarse-grained, interlocking crystals	Igneous - Granite
Sample B	Pyroxene, plagioclase	Fine-grained, dark	Igneous - Basalt
Sample C	Quartz, feldspar grains	Grained, cemented	Sedimentary - Sandstone
Sample D	Calcite	Crystalline	Sedimentary - Limestone
Sample E	Mica, quartz	Foliated, fine-grained	Metamorphic - Slate
Sample F	Feldspar, quartz, mica	Foliated, banded	Metamorphic - Gneiss

Discussion

The mineral identification exercise demonstrated the importance of using multiple physical properties for accurate identification. For example, while color can be variable (quartz can be many colors), hardness and cleavage are more reliable diagnostic features.

Key Observations:

- Hardness tests were particularly useful for distinguishing similar-looking minerals
- The streak test was crucial for identifying metallic minerals (pyrite vs. gold)
- Cleavage patterns provided strong diagnostic information (e.g., calcite's rhombohedral cleavage)
- Texture was the most important factor in rock classification

Challenges:

- Some minerals had variable appearance (feldspar color variations)
- Fine-grained rocks required hand lens for mineral identification
- Distinguishing between similar metamorphic rocks (slate vs. phyllite) required careful observation

Conclusion

This laboratory exercise successfully taught fundamental skills in mineral and rock identification. By systematically observing physical properties, we were able to identify common minerals and classify rocks according to their origin. These skills form the foundation for field geology and understanding Earth's processes.

The relationship between minerals and rocks became clear: rocks are composed of minerals, and the mineral composition and texture reflect the rock's formation process. Igneous rocks showed interlocking crystals, sedimentary rocks showed cemented grains or chemical precipitates, and metamorphic rocks showed foliation or recrystallization textures.